Experiment (4)

FRICTION FORCES

4.1 Introduction

We all know from our experience that when a body slides over a rough surface a “frictional force” often impedes the motion. Why is there friction when two surfaces slide opposite each other? When viewed at the microscopic level, friction is actually a very complex phenomenon. However, we can quantify and measure frictional forces on a macroscopic scale. In this lab, we will try to document the factors that determine the frictional force on a macroscopic object.

4.2 Objectives

- Investigate frictional forces (static and kinetic).
- Investigate the effect of the weight of the body on the frictional force. .
- Investigate the effect of the contact area of the body on the frictional force.
- Investigate the effect of the type of contact surface on the frictional force.
- Determine the coefficient of static friction and the coefficient of kinetic friction at the surface between two bodies.

4.3 Theory

When a mass block is pulled across a surface with a force T, a resisting frictional force is established.

If the pulling force T is not enough to move the body, the resisting frictional force is called the '*static*' frictional force, f_s . Since the body is not accelerating , the static frictional force in this case is equal to the pulling force.

$$f_s = T \tag{4.1}$$

As the pulling force T is gradually increased, the static frictional force also increases until the body **starts to move.** At this stage the static frictional force reaches its maximum value $f_{s,max}$ which is found to be proportional to the normal force n and is given by,

$$f_{s,max} = \mu_s n \tag{4.2}$$

where μ_s is called the coefficient of the static friction.

As soon as the body starts to move the frictional force usually decreases to a value f_k which is called the '*kinetic*' frictional force given by,

$$f_k = \mu_k n \tag{4.3}$$

where μ_k is called the coefficient of the kinetic friction.

Consider a block of mass m moving on an inclined plane of inclination angle α to the horizontal as shown in Figure 4.1. If the angle α is increased until the block starts to move at angle α_c, then, as shown in the figure, we obtain Equation (4.2) for the maximum frictional force

$$f_{s,max} = mg \sin\alpha_c = \mu_s n = \mu_s mg \cos\alpha_c \tag{4.4}$$

Combining Equation (4.2) with Equation (4.2) , the coefficient of the static friction is given by,

$$\mu_s = \frac{mg\sin\alpha_c}{mg\cos\alpha_c} = \tan\alpha_c \qquad (4.5)$$

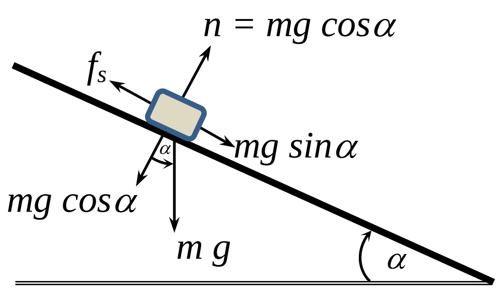

Figure 4.1: Forces acting on a body on an inclined surface.

4.4 Setup

The items used in this experiment are shown in Figure 4.2

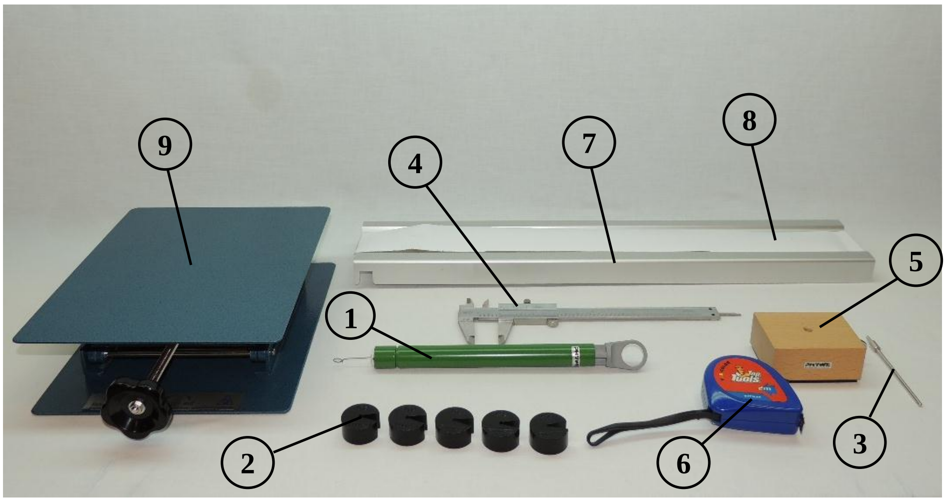

Figure 4.2: Experiment components.

1 Spring balance
2 Slotted masses, 50 gm
3 Holding pin
4 Vernier caliper
5 Frictional block with a rubber surface
6 Measuring tape
7 Frictional track
8 Drawing paper with rough surface
9 Lab jack

4.5 Procedure

You should calibrate the spring balance (1) horizontally or vertically before starting the measurements as shown in Figure 4.3 and Figure 4.4, depending on how it will be oriented during the measurement.

Figure 4.3: Horizontal calibration.

Figure 4.4: Vertical calibration.

4.5.1 Task (1): The effect of the weight of the block on the frictional force.

1- Measure the weight F_g of the frictional block (5) (including the holding pin (3)) by using the spring balance (1) as shown in Figure 4.5 and record the value in the lab report. *Make sure to calibrate the spring balance <u>vertically</u> before performing this step.*

2- Place the frictional block on a drawing paper with the rough surface (8), the rubber surface should touch the paper and the holding pin should face upwards.

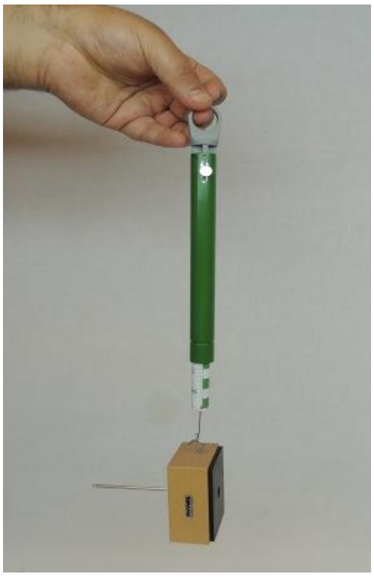

Figure 4.5: Measuring the weight of the frictional block.

3- Pull the frictional block slowly using the spring balance as shown in Figure 4.6. Just before it starts moving, read the value of the spring balance. This value represents the maximum static frictional force. *Make sure to calibrate the spring balance <u>horizontally</u> before performing this step*. Record this value in the Lab report.

Figure 4.6: Pulling the frictional block and measuring the frictional force.

The pulling force direction should be in the same direction of motion as shown in Figure 4.8 so the angle between the pulling force from the spring and the direction of motion = zero, ***why?***

Figure 4.7: Correct and incorrect angles of the pulling force.

4- Repeat steps 3 several times and each time add 50 *gm* slotted mass to the holding pin as shown in Figure 4.8.

5- Plot the relation between the frictional force on the vertical axis and the total weight of the frictional block on the horizontal axis.

Figure 4.8: Adding slotted mass to the frictional block.

6- Verify that the relation is linear and draw the best-fit line which passes through the points.
7- From the graph, calculate the slope of the line, use this value to get the coefficient of static friction μ_s.

For kinetic frictional coefficient :

Take one measurement as a demonstration as shown in steps 8 and 9 then show your work for your TA.

8- Repeat steps from 1 to 7 but in this case pull the frictional block slowly by using the spring balance to gradually increase the force as shown in Figure 4.6 above. When it starts moving, continue in pulling gently such that the block moves with a uniform velocity and read the pulling force value of the spring. The reading on the spring balance represents the kinetic frictional force. Record this value in the Lab report.
9- Use the recorded value of the kinetic frictional force to calculate the coefficient of kinetic friction μ_k.

4.5.2 Task (2): The effect of the type of the contact surface of the block on the frictional force.

1- Use the same setup that you used in Task (1).
2- Place the frictional block (5) on **<u>the bench surface directly</u>** with the rubber surface touching the bench surface and the holding pin facing upwards.
3- Repeat the steps 3 to 9 from Task (1). Record the values of the maximum static frictional force and the kinetic frictional force and find the frictional coefficients in this case in your lab report.
4- Compare the results obtained in this case with the results in Task (1). Write your comments on the effect of the type of the contact surface on the frictional force and the frictional coefficients.

4.5.3 Task (3): The effect of the area of the contact surface on the frictional force.

1- Use the vernier caliper (4) to measure the area of both the wide wood face and the side wood face of the block.
2- Place the frictional block (5) on the drawing paper with a rough surface (8), such that the wide wood face surface touches the sheet and the holding pin (3) is facing upwards as shown in Figure 4.9.

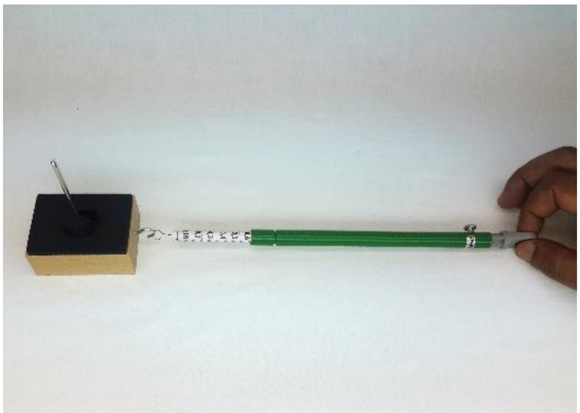

Figure 4.9: The wide wood surface touching the sheet.

3- Add 50 *gm* slotted mass (2) to the holding pin.
4- Pull the frictional block slowly by using the spring balance. Just before it starts moving, read the value of the spring balance. This value represents the maximum static frictional force. When it starts moving, continue in pulling gently such that the block moves with a uniform velocity and read the pulling force value of the spring. The reading on the spring balance represents the kinetic frictional force. Record these values in your lab report.
5- Repeat step 4 several times and each time add 50 *gm* slotted mass to the holding pin.
6- Use the values of the maximum static frictional force and the kinetic frictional force to clculate the frictional coefficients in this case as done in Task (1).
7- Repeat the last steps, but with the small side wood face touching the drawing paper with the rough surface as shown in Figure 4.10.

8- Compare the results obtained in both cases and write your comments about the effect of the area of the contact surface on the frictional force and the frictional coefficients.

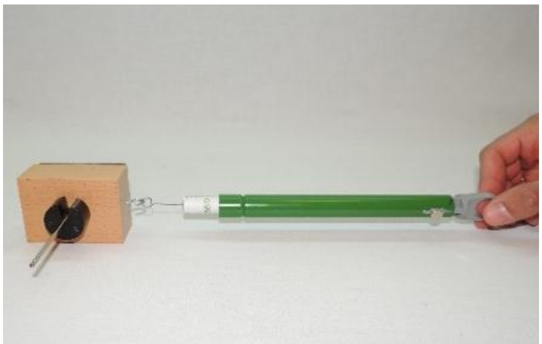

Figure 4.10: frictional force of the side face of the block.

4.5.4 Task (4): Measurement of the coefficient of the static

$$tan\ \alpha_c = \frac{opposite}{adjecent} = \frac{H}{L} \quad (4.6)$$

friction μ_s using the method of motion on inclined plane.

1- Setup the device as shown in Figure 4.11.

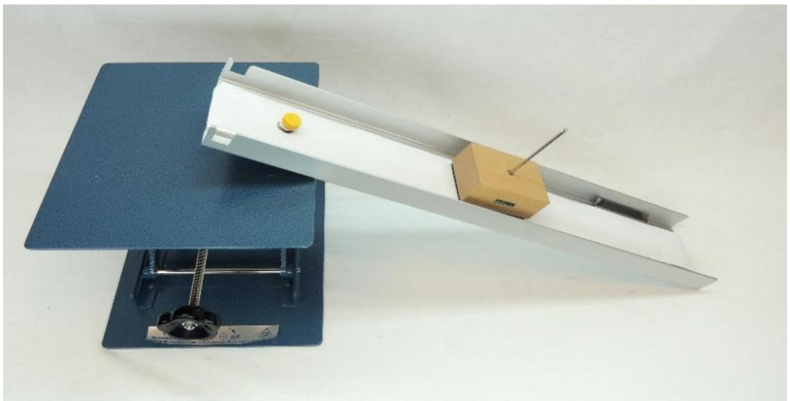

Figure 4.11: Setup of inclined plane.

2- Place the frictional block with its rubber face on the inclined track (7).

3- Use the lab jack (9) knob to increase the angle of inclination α gradually as shown in Figure 4.12 until the critical inclination angle α_c is reached *(the angle at which the block starts to slide).*

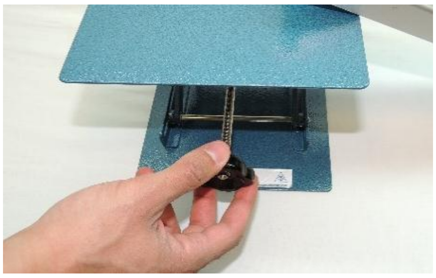

Figure 4.12: Increasing the angle of inclination.

4- Use the measuring tape (6) to measure the tangent of the inclination angle as shown in Figure 4.13.
where,

5- Record the value of $tan\ \alpha_c$ in your lab report.

6- Repeat steps 2 to 5 several times for error analysis then calculate the average value of $tan\alpha_c$. This value represents the static frictional coefficient μ_s.

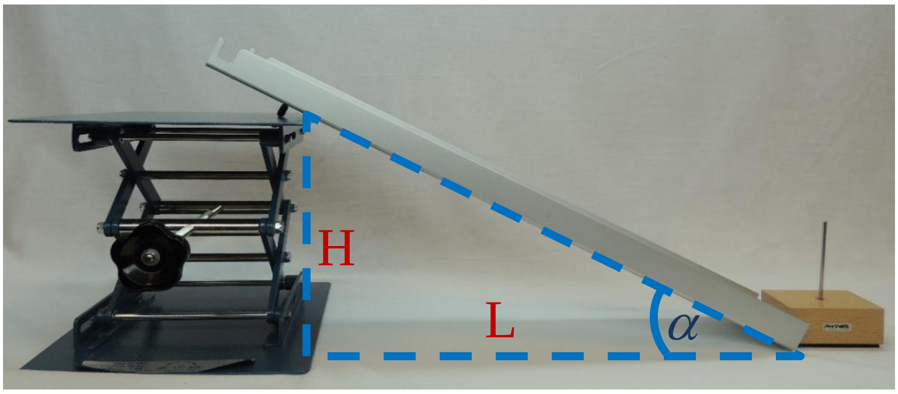

Figure 4.13: Measuring the angle of inclination.

7- Repeat steps 2 to 5 while adding masses to the frictional block and calculate the average value of $tan\ \alpha_c$ for each added mass. This result represents the value of the static frictional coefficient μ_s.

8- Compare your result in this case with the result you got in Task (1) for the same area and type of the contact surface.